Polluted Black Hole Companions: Spallogenic Lithium from Hot Accretion Flows with 3-D Weighted Monte Carlo Models

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Polluted white dwarfs record the bodies they accrete. Doyle et al. (2021) went one step further and read a nuclear signature in that pollution: the beryllium excess of GALEX J2339−0424 requires the accreted ice to have been bombarded by MeV protons in a giant planet's radiation belt. We apply the same reasoning to black holes. The hot, two-temperature accretion flow around a quiescent black hole is a far more intense radiation belt, with ion temperatures of 4–20 MeV inside 20 Schwarzschild radii. Spallation there makes lithium from helium and from carbon, nitrogen and oxygen, and beryllium from oxygen, and a small fraction of the processed gas can escape and land on the companion star. We extend the SERPENS weighted Monte Carlo code with an absorbing boundary at the innermost stable circular orbit and follow ions evaporated from the hot flow, comparing REBOUNDx post-Newtonian dynamics against exact Schwarzschild geodesics. About 1.5% of the evaporated ions escape. Exact general relativity and Newtonian gravity agree on this number, while the first post-Newtonian approximation overestimates it by a factor of 3.7 at $10\,r_s$. Combining the escape fraction with Roche-lobe geometry, we find that 0.9–1.5×10^{-4} of the accreted gas reaches the companion in A0620−00, XTE J1118+480 and V404 Cyg. Yi & Narayan (1997) needed between 10^{-6} and 10^{-4} for this mechanism to work but had to leave the number free. With accretion rates taken from published spectral fits and the observed outburst histories, the predicted surface lithium matches the measurements in A0620−00 and V404 Cyg and respects the upper limit in XTE J1118+480, for lithium depletion times of 4×10^7–3×10^8 yr that follow the expected order for a subgiant and a K dwarf. For the wide, dormant Gaia black holes the delivered lithium is at least ten orders of magnitude below detectability, consistent with the normal lithium of the Gaia BH1 companion and the absence of anomalies in Gaia BH2 and BH3. The same model predicts that the V404 Cyg companion carries about five times the solar beryllium abundance, which near-UV spectroscopy can test.

PACS numbers:

I. INTRODUCTION

White dwarfs are the cleanest chemical record of planetary material we have. Their surface gravity is high enough that metals sink out of the photosphere in days to millions of years, so any metal line in a white dwarf spectrum is material that arrived recently. Doyle, Desch & Young [1] used this record to argue for icy exomoons. The polluted white dwarf GALEX J2339−0424 carries roughly 200–500 times more beryllium (relative to iron) than CI chondrites. Beryllium is not made in stars in that quantity. It is made by spallation, when energetic protons break heavier nuclei such as oxygen into lighter fragments. Doyle et al. showed that the required number ratio $n_{\rm Be}/n_{\rm O}\sim10^{-5}$ follows from a proton fluence of $\sim10^{21}$ cm^{-2}, which ice gathers in about 10^7 yr inside a Jupiter-like radiation belt. The beryllium is a nuclear fingerprint of where the accreted body spent its time.

Our group developed SERPENS to follow trace material lost from volcanic exo-Ios into the environments of close-in giant planets [2, 3]. In this paper we turn the polluted white dwarf argument toward black holes. A black hole has no photosphere to hold a record, but its companion star does. And the environment near a quiescent black hole is a much harsher radiation belt than any planetary magnetosphere. At accretion rates well below Eddington, the inner flow is an advection-dominated accretion flow (ADAF) in which the ions stay near the virial temperature [4–6]. Inside $20\,r_s$ the ion temperature is 4–20 MeV, and the tail of the distribution reaches the thresholds for $\alpha+\alpha\rightarrow{}^{6,7}$Li and for proton spallation of carbon, nitrogen and oxygen.

The companions of several black hole X-ray transients do show the lithium excess this picture suggests. Martín et al. [7, 8] found strong Li I 6708 Å absorption in V404 Cyg, A0620−00, Nova Mus 1991 and the neutron star system Cen X-4, and lithium has since been detected in four black hole companions [9]. The K-type secondary of A0620−00 has $A(\mathrm{Li})=2.31\pm0.21$, compared with ≤0.35 in field stars of the same mass [10]. Stars this cool destroy their surface lithium through convective mixing during the pre-main sequence and main sequence [10]. Yi & Narayan [11] proposed that the lithium is made by α–α spallation in the hot flow and carried to the companion. Their estimate worked if between 10^{-6} and 10^{-4} of the accreted mass ends up on the secondary after processing, but they had no way to compute that fraction and treated it as a free parameter. They also pointed out that white dwarf accretion flows never get hot enough for this, so the lithium is a signature of a neutron star or black hole.

That fraction is a test-particle problem, and SERPENS is

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a test-particle code. Here we compute it. We launch ions from the hot flow with the SERPENS thermal evaporation mode, follow them through the strong-field region with an absorbing inner boundary at the innermost stable circular orbit (ISCO), tally the proton fluence each ion receives, and ask how much of the processed gas lands on the companion. We then compare the predicted surface lithium with the measured values in A0620−00, XTE J1118+480 and V404 Cyg, and with the three dormant Gaia black holes. Their companions have been analysed in detail. The Gaia BH1 G dwarf has $A(\mathrm{Li}) = 2.3 \pm 0.1$, which El-Badry et al. [12] find unremarkable for a solar-type star older than the Hyades, and its abundance pattern is typical of the thin disk. The Gaia BH2 red giant is α-enhanced, $[\alpha/\mathrm{Fe}] = +0.26 \pm 0.05$, but otherwise normal, and shows no lithium [13]. The very metal-poor Gaia BH3 giant shows no chemical peculiarity [14, 15].

Figure 1 shows the setup. Section II sets out the spallation physics and the relativistic length scales. Section III describes the SERPENS extension, the hot-flow model and the delivery calculation. Section IV presents the dynamics, the yields and the comparison with observations. We discuss the uncertainties in Section V and summarise in Section VI. The integrator benchmark from our earlier GR tests is in Appendix A.

II. SPALLATION NEAR A BLACK HOLE

A. The polluted white dwarf argument

For a target nucleus exposed to a fluence F of projectiles above the reaction threshold, the number of product nuclei per target grows as

$$\frac{n_{\mathrm{Be}}}{n_{\mathrm{O}}} = \sigma_{\mathrm{Be}} F, \qquad F = \int \Phi_{>}\, dt, \tag{1}$$

as long as $\sigma F \ll 1$. Here $\Phi_{>}$ is the flux of protons above ~ 10 MeV and $\sigma_{\mathrm{Be}} \approx 10^{-26}$ cm^2 is the $^{16}\mathrm{O}(p,x)^{9}\mathrm{Be}$ cross section [1]. Equation (1) is the relation Doyle et al. used for GALEX J2339−0424. We apply it unchanged; only the source of the protons is different.

B. The hot accretion flow as a radiation belt

In a two-temperature ADAF the ions are heated by viscous dissipation and lose energy to the electrons only through slow Coulomb collisions, so they stay close to the virial temperature [4, 5]. We write

$$kT_i(r) = \theta_i \frac{m_p c^2}{x}, \qquad x \equiv r/r_s, \tag{2}$$

with $\theta_i = 0.55/6$, i.e. 55% of the virial value $m_p c^2/(6x)$. This gives $kT_i \approx 86\,\mathrm{MeV}/x$, or 8.6 MeV at $10\,r_s$. The flow falls in at $v_r = c_1\,\alpha\, v_K$ with $v_K = (GM/r)^{1/2} = c/\sqrt{2x}$, and we take $\alpha = 0.1$ and $c_1 = 0.5$ [5]. Mass conservation then fixes the ion density,

$$n(r) = \frac{\dot{M}}{4\pi r^2\, v_r\, m_p}, \qquad \dot{M} = \dot{m}\dot{M}_{\mathrm{Edd}}, \tag{3}$$

where $\dot{M}_{\mathrm{Edd}} = L_{\mathrm{Edd}}/(0.1\,c^2)$ and $L_{\mathrm{Edd}} = 4\pi GMm_p c/\sigma_T$. The flux of ions above a threshold energy E_{th} in a Maxwellian at T_i is

$$\Phi_{>}(r) = n_j(r)\, \frac{\Gamma(3/2, E_{\mathrm{th}}/kT_i)}{\Gamma(3/2)}\, \bar{v}_{>}, \tag{4}$$

where n_j is the density of species j ($n_\alpha = 0.085\,n$), $\Gamma(a,y)$ is the upper incomplete gamma function and $\bar{v}_{>}$ is the mean speed of the ions above threshold. We use $E_{\mathrm{th}} = 10$ MeV for protons on oxygen and $E_{\mathrm{th}} = 35$ MeV (laboratory energy) for $\alpha + \alpha \to \mathrm{Li}$.

Because $n \propto \dot{m}/M$ and every timescale scales with M, the fluence a parcel collects is independent of the black hole mass and proportional to $\dot{m}$. The dose problem therefore has one parameter, the Eddington-scaled accretion rate.

C. Relativistic length scales

Two radii bound the region that matters for a non-spinning black hole. The horizon sits at $r_s = 2GM/c^2$ (29.5 km for $10\,M_\odot$) and the ISCO at $r_{\mathrm{ISCO}} = 3\,r_s$ [16]. An ion that crosses the ISCO on an inward trajectory cannot return on a stable orbit, so SERPENS removes it there and counts it as accreted. All of the spallation in our model happens between 3 and $\sim 30\,r_s$, where the escape speed is 0.2–0.6c. This is the regime where the choice of gravity model has to be checked rather than assumed, and we do so in Section IV A.

D. Is a test-particle treatment justified?

SERPENS is collisionless. Fluid dynamics applies when the Knudsen number $\mathrm{Kn} = \lambda_{\mathrm{mfp}}/L \ll 1$, and a kinetic treatment is needed when $\mathrm{Kn} \gtrsim 1$ [17]. ADAFs are in the second regime: the Coulomb collision time is longer than the inflow time, which is why the ions and electrons stay at different temperatures [4, 6]. What our test particles leave out is the magnetic field, which ties the ions to the flow on gyration scales far below r_s. We therefore treat each ion's ballistic escape as a single evaporation attempt and regard the resulting escape fraction as an estimate of the processed outflow, not a solution of the MHD problem. Section V returns to this point.

III. METHODS

A. SERPENS

Meyer zu Westram et al. [3] describe SERPENS, a 3-D weighted Monte Carlo code that follows superparticles launched from a source body. Each superparticle carries a species, a launch velocity drawn from a physical distribution and a weight set by the source mass-loss rate. The trajectories are integrated with REBOUND [18], and particles that collide with a massive body are removed. In the thermal evaporation

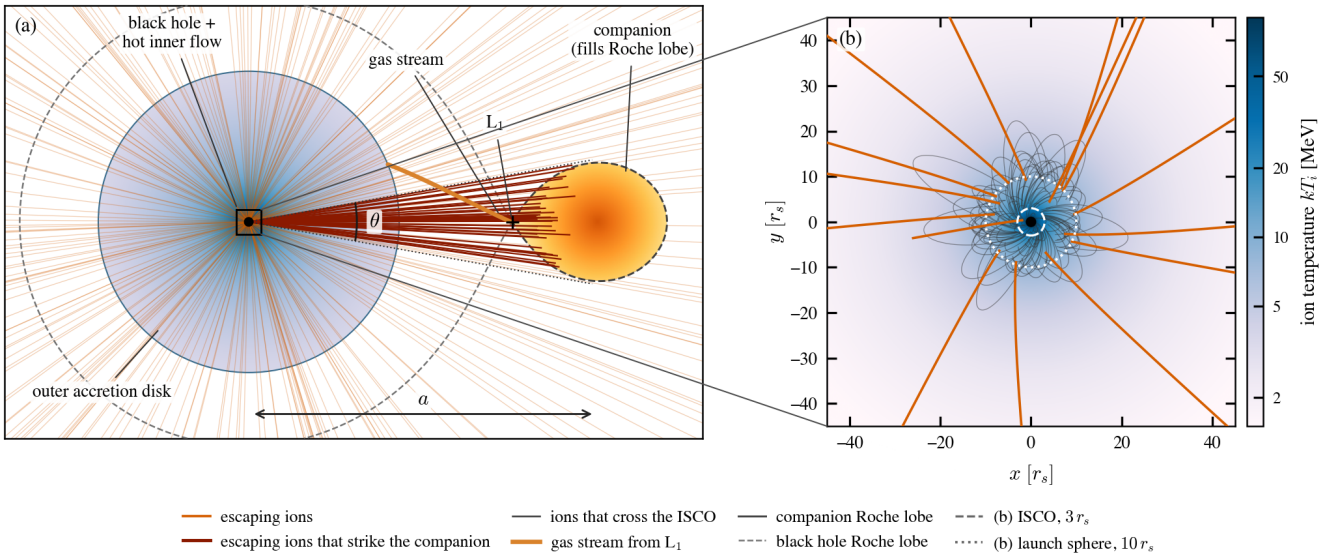


FIG. 1: Geometry of the problem, drawn to scale for A0620−00 ($q = M_2/M_{\rm BH} = 0.061$) in the frame co-rotating with the binary. *(a)* The K-dwarf companion fills its Roche lobe (solid contour through the inner Lagrange point L_1; the dashed contour is the black hole's lobe) and feeds the accretion disk, truncated at $0.43\,a$, through a ballistic stream from L_1. Thin lines are the projected directions of the 3230 ions that escape the hot flow out of 2×10^5 launches at $10\,r_s$ (exact Schwarzschild classification); the 42 that leave within the angle $\theta = R_L/a = 0.18$ of the companion strike it (dark red). *(b)* The hot flow around the black hole: SERPENS trajectories of ions evaporated at $10\,r_s$ (as in Figure 2) over the ion temperature of equation (2). The box in (a) is not to scale; panel (b) spans $3\times10^{-4}\,a$. SERPENS integrates the region in (b), and the fraction delivered to the companion follows from the geometry in (a) through equation (10).

mode, launch speeds follow a Maxwell–Boltzmann distribution,

$$f_{\rm th}(v) = \sqrt{\frac{2}{\pi}}\,\frac{v^2}{v_{\rm sc}^3}\exp\left(-\frac{v^2}{2v_{\rm sc}^2}\right), \qquad v_{\rm sc} = \sqrt{\frac{k_B T}{m}}. \tag{5}$$

B. Launching ions from the hot flow

We treat a sphere of radius r_0 around the black hole as the source surface. Positions are drawn uniformly on that sphere with the SERPENS position sampler, and each ion leaves radially with a speed drawn from equation (5) at $T = T_i(r_0)$ and $m = m_p$. On top of this we add the rotation of the flow, $\Omega = 0.4\,\Omega_K$ about the orbital axis [5]. We run r_0 from 4 to $50\,r_s$; most of the lithium is made inside $20\,r_s$, so we quote results at $r_0 = 10\,r_s$ and use 5–$20\,r_s$ as the range. Units are r_s for length and r_s/c for time, which makes the dynamics mass-independent.

C. Gravity near the black hole

We integrate the ions with the adaptive IAS15 integrator [19] in batches of 150–400 particles; REBOUND's collision search scales as N^2 even for test particles, so small batches run in parallel are much faster than one large run. The black hole is a massive body with a collision radius equal to $r_{\rm ISCO}$, and the `line` collision search catches ions that cross the ISCO between steps. We run each launch radius three ways:

1. Newtonian gravity in SERPENS;
2. SERPENS with the REBOUNDx `gr` force [20], the single-source 1PN acceleration of Anderson et al. [21],

$$\mathbf{a} = -\frac{GM}{r^2}\hat{\mathbf{r}} + \frac{GM}{c^2 r^2}\left[\left(\frac{4GM}{r} - v^2\right)\hat{\mathbf{r}} + 4\,(\hat{\mathbf{r}}\cdot\mathbf{v})\,\mathbf{v}\right]; \tag{6}$$

3. exact Schwarzschild geodesics, evaluated analytically.

Around an isolated black hole the third case needs no integration. Taking the launch speed as measured by a static observer, each ion has conserved specific energy and angular momentum (in units with $c = r_s = 1$)

$$\mathscr{E} = \gamma\sqrt{1 - 1/x}, \qquad \mathscr{L} = \gamma x v_t, \tag{7}$$

and moves in the effective potential

$$V^2(x) = \left(1 - \frac{1}{x}\right)\left(1 + \frac{\mathscr{L}^2}{x^2}\right) \tag{8}$$

[22]. An ion escapes if $\mathscr{E} \geq 1$ and $\mathscr{E}^2$ exceeds V^2 everywhere outside its position; it plunges if $\mathscr{E}^2$ exceeds V^2 everywhere between 3 and x on its way in; otherwise it stays bound to the flow. The Newtonian version replaces equations (7)–(8) with $E = v^2/2 - 1/(2x)$ and $V = -1/(2x) + \mathscr{L}^2/(2x^2)$. Ions in the Monte Carlo runs that reach $10^3\,r_s$ are classified by their energy there; ions still inside at $t = 2\times10^4\,r_s/c$ are classified at that time.

TABLE I: Fiducial hot-flow and spallation parameters.

Quantity	Value	Source
Ion temperature kT_i	$0.55\times m_p c^2/(6x)$	[5]
Viscosity, c_1	$\alpha=0.1$, $c_1=0.5$	[5]
Rotation Ω/Ω_K	0.4	[5]
He/H, O/H (number)	0.085, 4.9×10^{-4}	[24]
$\sigma_{\rm Be}$ (p+O, >10 MeV)	1×10^{-26} cm^2	[1]
$\sigma_{\rm Li}$ (α+α, >35 MeV)	5×10^{-26} cm^2	[23]
$\sigma_{\rm Li}^{\rm CNO}$ (p+CNO, >10 MeV)	1×10^{-26} cm^2	[1, 23]
CNO/H (number)	8.3×10^{-4}	[24]
σ_d (destruction by p)	2×10^{-25} cm^2	representative
Li depletion time $\tau_{\rm dep}$	10^7–10^9 yr	[11]

D. Fluence and yields

A parcel collects fluence twice: while it flows in from $10^3\,r_s$ to r_0, and, if it escapes, along its way out. For the inflow we integrate equation (4) along $dt=dr/v_r$. For the outflow we accumulate $\Phi_>(r)\,\Delta t$ at every IAS15 step through a REBOUND heartbeat. The outflow contributes about 3% of the total, so the inflow sets the yield.

We evolve the light-element abundances along the inflow with production and proton-induced destruction,

$$\begin{aligned} d\!\left(\frac{\rm Li}{\rm H}\right) &= \frac{n_\alpha}{n_{\rm H}}\,\sigma_{\rm Li}\,dF_\alpha + \frac{n_{\rm CNO}}{n_{\rm H}}\,\sigma_{\rm Li}^{\rm CNO}\,dF_p - \sigma_d\frac{\rm Li}{\rm H}\,dF_p,\\ d\!\left(\frac{\rm Be}{\rm O_0}\right) &= \sigma_{\rm Be}\,\frac{\rm O}{\rm O_0}\,dF_p - \sigma_d\frac{\rm Be}{\rm O_0}\,dF_p,\\ d\!\left(\frac{\rm O}{\rm O_0}\right) &= -\sigma_d\,\frac{\rm O}{\rm O_0}\,dF_p, \end{aligned} \tag{9}$$

which reduce to equation (1) when $\sigma_d F_p \ll 1$. The cross sections are listed in Table I. They are representative values near the cross-section peaks [1, 23], and Section V gives the sensitivity to them. The two lithium channels contribute almost equally. Fast α particles are rarer than fast protons, because their threshold sits deeper in the Maxwellian tail, and this offsets the larger abundance of helium. We include both; leaving out the CNO channel would lower every lithium prediction by 0.3 dex.

E. Delivery to the companion

The donors in black hole X-ray transients fill their Roche lobes, so each subtends an angle $\theta=R_L/a$ seen from the black hole (Figure 1a), with R_L/a from Eggleton's formula [25]. Averaged over the orbit, the companion sweeps a band of half-width θ around the orbital plane and covers a fraction $\theta/4$ of that band at any time. The fraction of the accreted gas that reaches the companion is then

$$f_{\rm int}=f_{\rm esc}\,g\,\frac{\theta^2}{4}, \tag{10}$$

where $f_{\rm esc}$ is the escape fraction from Section IV A and g is the density of escaping ions near the orbital plane relative to an isotropic outflow, measured from the launch directions of the escaping ions. Gravitational focusing by the companion is negligible for ions leaving at $\gtrsim 0.1c$.

If the delivered lithium mixes into a surface convection zone of mass $M_{\rm conv}$ and is destroyed there on a timescale $\tau_{\rm dep}$, the steady-state surface abundance is

$$\left(\frac{\rm Li}{\rm H}\right)_\star=\frac{\dot M_{\rm Edd}\,\langle\dot m\,y_{\rm Li}(\dot m)\rangle\,f_{\rm int}\,\tau_{\rm dep}}{M_{\rm conv}}, \tag{11}$$

with $A({\rm Li})=12+\log_{10}({\rm Li/H})_\star$ and $y_{\rm Li}(\dot m)$ the Li/H of a processed parcel. Because $y_{\rm Li}\propto\dot m$, the delivered lithium scales as $\dot m^2$, which makes the accretion history the main input after the dynamics.

F. Accretion rates and outburst cycles

For A0620−00 we adopt the ADAF fit of Narayan, McClintock & Yi [26], $\dot m=4\times10^{-4}$. For V404 Cyg, Narayan, Barret & McClintock [27] fit $\dot M=1.22\times10^{-9}M_\odot$ yr^{-1} for $M=12M_\odot$ at 3.5 kpc. Rescaled to $M=9M_\odot$ [28] and the parallax distance of 2.39 kpc [29], this is $\dot m=5.0\times10^{-3}$. XTE J1118+480 has no published fit, so we tie the efficiency to the A0620−00 fit at $L/L_{\rm Edd}=8.5\times10^{-9}$ [30] and to the hard-to-soft transition near 2% of Eddington [31]:

$$\frac{L}{L_{\rm Edd}}=0.02\left(\frac{\dot m}{0.02}\right)^P,\qquad P=3.75. \tag{12}$$

Applied to V404 Cyg, equation (12) gives $\dot m=1.1\times10^{-3}$, a factor of 4.6 below the direct fit, which indicates how well the relation does. For XTE J1118+480 at $L_X\approx4\times10^{-9}L_{\rm Edd}$ [32] it gives $\dot m=3.3\times10^{-4}$. The simpler linear efficiency we used for Gaia BH1 in an earlier draft would put all three systems 40–85 times lower in $\dot m$; that prescription is too efficient at these rates and we no longer use it.

The transients also spend time at higher $\dot m$ during outbursts. We add the hard-state parts of each outburst, below 2% of Eddington where the hot flow survives [31], with luminosity decaying as $L=L_0\,e^{-t/\tau_e}$ and $\dot m(t)$ from equation (12):

$$\langle\dot m\,y\rangle=\dot m_q\,y(\dot m_q)+\frac{1}{T_{\rm rec}}\sum_{\rm segments}\int\dot m(t)\,y(\dot m(t))\,dt. \tag{13}$$

A0620−00 erupted in 1917 and 1975 ($T_{\rm rec}=57.8$ yr) and its 1975 outburst decayed with $\tau_e=29$ d [33, 34]. V404 Cyg has recurred roughly every 26 yr, most recently in 1989 and 2015 [35]; we use the generic $\tau_e=30$ d [34]. For both we include the decline through the hard state and a 3 d hard-state rise. XTE J1118+480 has had only hard-state outbursts, in 2000 (peak 1.2×10^{36} erg s^{-1}) and 2005 (peak 2×10^{35} erg s^{-1}) [36]; we model them as a 100 d plateau and a 5 d plateau with 30 d and 5 d decays, and count both once per 25 yr.

Table II lists the adopted system parameters. The convective-zone masses are our estimates for the spectral types and carry at least a factor of two uncertainty; they enter equation (11) linearly.

TABLE II: Adopted parameters and observed lithium. $\theta = R_L/a$ for the Roche-lobe-filling donors and R_2/a for the detached Gaia systems; for Gaia BH3 a is the relative separation, $a_1(M_{\rm BH}+M_2)/M_{\rm BH}$, from the published $a_1 = 16.17$ AU. The XTE J1118+480 donor mass follows from $q = 0.027$ [37]; its lithium is an upper limit.

System	$M_{\rm BH}$ [$M_\odot$]	M_2 [$M_\odot$]	$P_{\rm orb}$	θ	$\dot m_q$	outbursts	$M_{\rm conv}$ [$M_\odot$]	observed A(Li)
A0620−00	6.6 [38]	0.40 [38]	7.75 h	0.178	4.0×10^{-4} [26]	1917, 1975	0.15	2.31 ± 0.21 [10]
XTE J1118+480	7.5 [39]	0.20 [37]	4.08 h	0.139	3.3×10^{-4} (eq. 12)	2000, 2005 (hard)	0.20	≤ 1.78 [37]
V404 Cyg	9.0 [28]	0.54 [40]	6.47 d	0.178	5.0×10^{-3} [27]	∼26 yr recurrence	0.30	2.70 ± 0.40 [9]
Gaia BH1	9.62 [12]	0.93 [12]	186 d	3.3×10^{-3}	7×10^{-12} (BHL)	none	0.03	2.3 ± 0.1, normal [12]
Gaia BH2	8.94 [13]	1.07 [13]	3.50 yr	7.3×10^{-3}	2×10^{-9} [13]	none	0.8	not detected (giant) [13]
Gaia BH3	32.7 [14]	0.76 [14]	11.6 yr	1.4×10^{-3}	10^{-8} [41]	none	0.5	no peculiarity [15]

G. The dormant Gaia black holes

The three Gaia black holes are detached and fed only by their companions' winds. For Gaia BH1 we use the SERPENS wind mode from our earlier work: a shifted Maxwellian with the Wood et al. [42, 43] mass-loss rate, $\dot M_w \approx 1.3\times10^{9}$ kg s^{-1}, and a Parker asymptotic speed of ≈ 830 km s^{-1} [44, 45]. The Bondi–Hoyle–Lyttleton rate [46, 47],

$$\dot M_{\rm BHL} = \pi r_{\rm BHL}^2 \rho_{\rm wind} v_{\rm rel}, \qquad r_{\rm BHL} = \frac{2GM_{\rm BH}}{v_{\rm rel}^2}, \tag{14}$$

gives $r_{\rm BHL} \approx 4\times10^{9}$ m and $\dot M_{\rm BHL} \approx 1\times10^{5}$ kg s^{-1} at the 1.4 AU separation, or $\dot m \approx 7\times10^{-12}$. Rodriguez et al. [48] estimate $\dot m \approx 5\times10^{-12}$ at the horizon, close to our value. For the two giants we take the published estimates. El-Badry et al. [13] reduce the Bondi–Hoyle–Lyttleton rate onto Gaia BH2 to $\dot m \approx 2\times10^{-9}$ at the horizon, and Cappelluti, Pacucci & Hasinger [41] bracket Gaia BH3 between 10^{-9} and 10^{-7}; we adopt 10^{-8}. All three systems are X-ray dark. The *Chandra* limits are 2.8×10^{29} erg s^{-1} for Gaia BH1 (0.5–7 keV) [48], $F_X < 5.1\times10^{-15}$ erg s^{-1} cm^{-2} or $L_X \lesssim 8\times10^{29}$ erg s^{-1} at 1.16 kpc for Gaia BH2 [13], and 2.1×10^{29} erg s^{-1} (2–10 keV) for Gaia BH3 [41]. Because the Li yield scales as $\dot m$ and the delivery as $\dot m^2$, a factor of ten in any of these rates moves the predicted lithium by two dex, which does not change our conclusions.

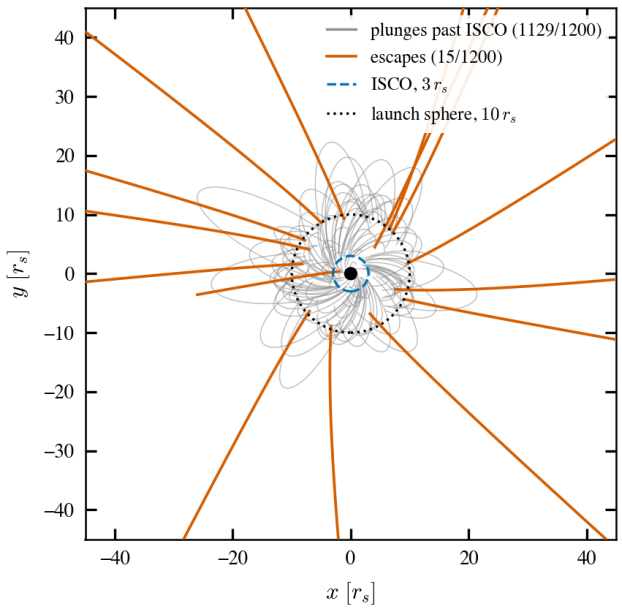


FIG. 2: SERPENS trajectories of 1200 ions evaporated from the hot accretion flow at $r_0 = 10\,r_s$ around a non-spinning black hole, projected onto the orbital plane. Grey: a random 120 of the 1129 ions that cross the ISCO (dashed blue) and are removed as accreted. Red: the 15 ions that escape. The black disk is the horizon and the dotted circle is the launch sphere. Newtonian gravity, IAS15 integrator.

IV. RESULTS

A. Fate of the evaporated ions

Figure 2 shows 1200 ions launched from $10\,r_s$. Almost all of them go up and fall straight back: the thermal speeds ($v_{\rm sc} \approx 0.1c$ at $10\,r_s$) are a third of the escape speed, and the flow's rotation gives the ions too little angular momentum to hold them off the ISCO. In this run 1129 ions crossed the ISCO, 15 escaped, and the remaining 56 were still on wide bound excursions at the end of the integration. The escapers are the high-speed tail of the Maxwellian, and they leave close to radially.

Figure 3 gives the escape fraction against launch radius. Exact Schwarzschild geodesics and Newtonian gravity agree at every radius, at 1.5–1.7%. This is not a coincidence. For radial motion the GR escape condition $\mathscr{E} \geq 1$ reduces to $v^2 \geq r_s/r = 2GM/(rc^2)$, the Newtonian condition, when v is the speed a static observer measures. The SERPENS Newtonian runs reproduce the analytic classification particle by particle: in a 2000-ion test at $10\,r_s$, 99.8% of the ions have the same outcome. The escape fraction barely changes with r_0 because kT_i and $v_{\rm esc}^2$ both scale as $1/r$, so the problem looks the same at every radius.

The REBOUNDx 1PN force does not. At $10\,r_s$ it lets 6.3% of the ions escape, 3.7 times the exact value; the excess falls to a factor of 1.7 at $20\,r_s$ and 1.2 at $50\,r_s$. Inside $10\,r_s$ the iterative 1PN solver stops converging and the integration stalls. The reason is visible in equation (6). For an ion moving radially outward the bracket becomes $4GM/r + 3v^2$, which points outward. At $10\,r_s$ and $v = 0.35c$ it cancels 57% of the Newtonian pull. The 1PN expansion assumes $v^2/c^2 \ll 1$, and the

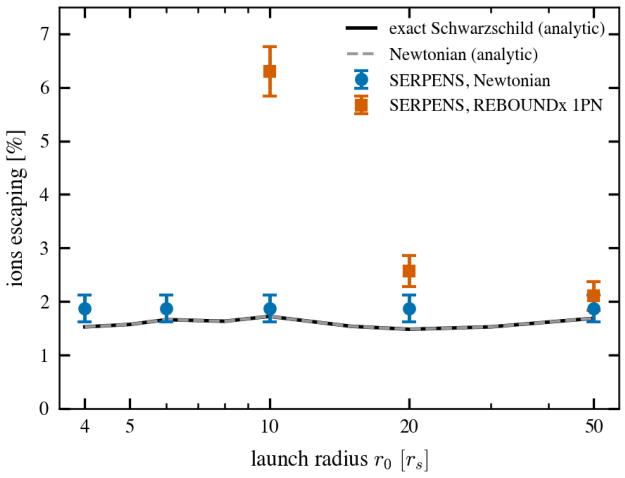


FIG. 3: Percentage of evaporated ions that escape, against launch radius. Lines: analytic classification of 20 000 launches with exact Schwarzschild geodesics (solid) and Newtonian gravity (dashed); the two coincide. Points: SERPENS Monte Carlo runs of 3000 ions with Newtonian gravity (circles) and with the REBOUNDx 1PN force of equation (6) (squares), with Poisson errors. The 1PN runs do not converge inside $10\,r_s$.

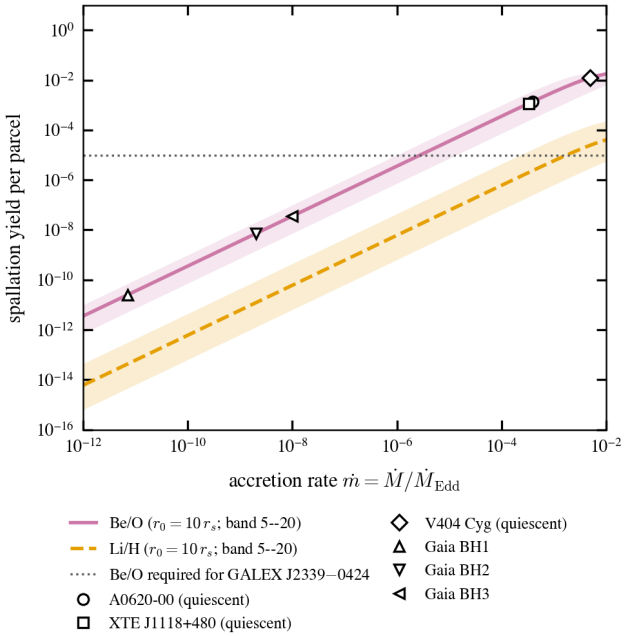


FIG. 4: Spallation yield per parcel against Eddington-scaled accretion rate: Be per initial O (solid) and Li per H (dashed), for $r_0 = 10\,r_s$ with shaded bands for $r_0 = 5$–$20\,r_s$. The dotted line is the Be/O that Doyle et al. [1] require for the polluted white dwarf GALEX J2339−0424. Markers show the quiescent accretion rates adopted for the six black holes.

escaping ions violate that assumption by construction. The 1PN treatment remains accurate where our earlier tests used it, at binary separations where $v/c \sim 10^{-3}$ (Appendix A), but it should not be used for ballistic motion inside $\sim 20\,r_s$. From here on we use the exact Schwarzschild classification, $f_{\rm esc} = 1.55\%$, with an equatorial enhancement $g = 1.22$ from the flow's rotation.

B. Spallation yields

Figure 4 shows the yield of a parcel that reaches r_0 as a function of accretion rate. Both yields are linear in $\dot{m}$ until destruction sets in near $\dot{m} \sim 10^{-2}$. A parcel takes about 40 s to fall from 10^3 to $10\,r_s$ around a $7\,M_\odot$ black hole. At the quiescent accretion rate of A0620−00 that is enough for a fluence of 1.4×10^{23} cm^{-2} above 10 MeV and a Be/O ratio of 1.4×10^{-3}. Doyle et al. needed 10^{-5} for GALEX J2339−0424, which ice in a Jovian radiation belt reaches in $\sim 10^7$ yr. The hot flow does the same work in under a minute, already at $\dot{m} \approx 3 \times 10^{-6}$.

C. Comparison with the observed lithium

Equation (10) gives $f_{\rm int} = 1.5 \times 10^{-4}$ for A0620−00 and V404 Cyg, which have nearly identical mass ratios, and 0.9×10^{-4} for XTE J1118+480, which has a smaller donor. All three lie at the upper end of the 10^{-6}–10^{-4} range that Yi & Narayan [11] needed. In their notation the escaping fraction was $F_{\rm esc} \approx 10^{-3}\,(10^8\,{\rm yr}/\tau_{\rm dep})$; our $f_{\rm esc} = 1.5 \times 10^{-2}$ corresponds to the short end of their depletion times. Figure 5 and Table III compare the predictions with the measurements.

With outbursts included, the predicted range contains the observed lithium in A0620−00 and V404 Cyg, and the computed $f_{\rm int}$ falls inside the required range for both. For XTE J1118+480 the measurement is an upper limit, $A({\rm Li}) \leq 1.78$, and the model respects it for $\tau_{\rm dep} \lesssim 10^8$ yr. The three systems need different depletion times: $\tau_{\rm dep} \approx 4 \times 10^7$ yr for the V404 Cyg subgiant and $\approx 3 \times 10^8$ yr for the A0620−00 K dwarf. This is the order expected from stellar structure. Yi & Narayan [11] quote depletion times of a few $\times 10^8$ yr for main-sequence K dwarfs and $\sim 5 \times 10^7$ yr for giants. How much the outbursts matter depends on the system. In V404 Cyg the quiescent flow already accretes at $\dot{m} = 5 \times 10^{-3}$ and the outbursts add only 0.03 dex. In A0620−00 the quiescent flow alone falls 0.2 dex short even at $\tau_{\rm dep} = 10^9$ yr, and the hard-state phases of the outbursts raise the prediction by 0.7 dex. We did not adjust any parameter to obtain this agreement: every input in Tables I and II was fixed before the comparison.

D. The dormant Gaia black holes

The Gaia black holes sit at the other extreme. Their accretion rates are four to nine orders of magnitude below those of the transients, so each parcel carries correspondingly less lithium (Figure 4). Their companions are also far away: θ is 3.3×10^{-3} for Gaia BH1, 7.3×10^{-3} for the larger Gaia BH2

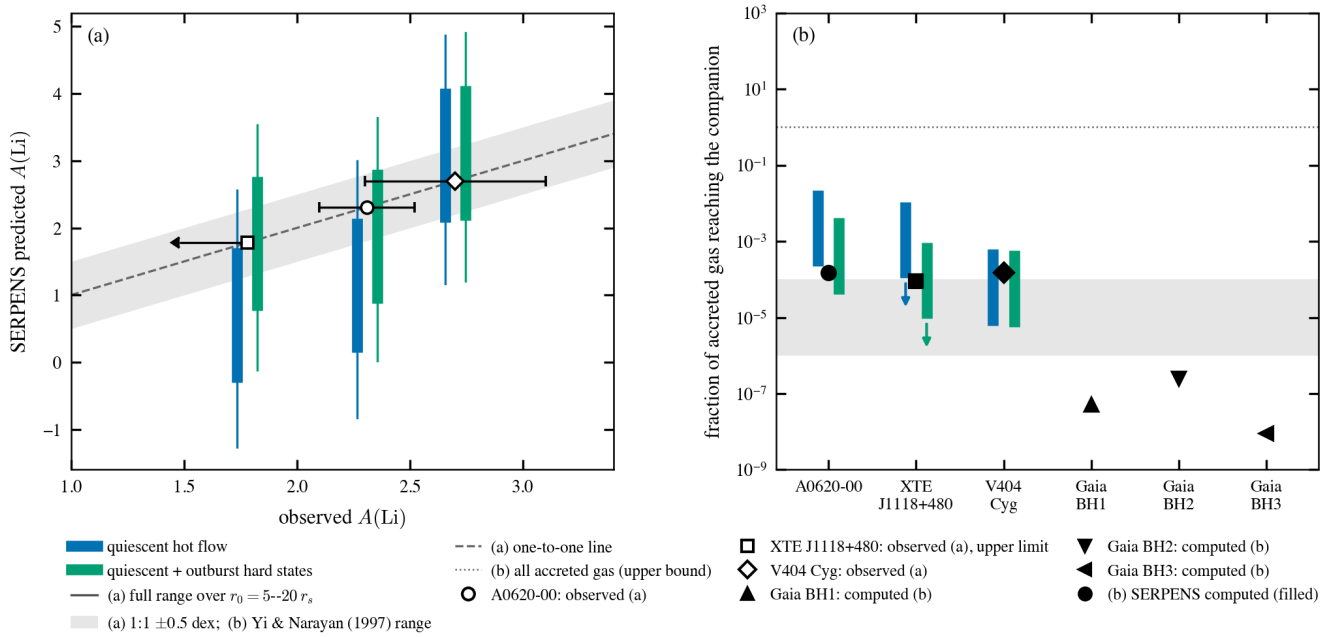


FIG. 5: SERPENS predictions against observations. *(a)* Predicted surface lithium of the companion against the measured value. Thick bars span depletion times $\tau_{\rm dep} = 10^7$–10^9 yr at $r_0 = 10\,r_s$; thin lines extend over $r_0 = 5$–$20\,r_s$. Blue: quiescent hot flow only. Green: quiescent flow plus the hard-state phases of the observed outbursts. Open symbols mark the observations on the one-to-one line; the arrow marks the XTE J1118+480 upper limit. The grey band is ±0.5 dex. *(b)* Fraction of the accreted gas that must reach the companion to reproduce the measured lithium (bars, same colours and $\tau_{\rm dep}$ range) and the fraction SERPENS computes from equation (10) (filled symbols). The grey band is the range required by Yi & Narayan [11]; the dotted line marks delivery of all the accreted gas. Arrows mark the XTE J1118+480 requirement, which is an upper limit. The Gaia black holes are shown for their computed fractions; none of their companions shows a lithium excess.

TABLE III: Predicted and observed companion lithium, and predicted beryllium, at $r_0 = 10\,r_s$. Ranges span $\tau_{\rm dep} = 10^7$–10^9 yr. $f_{\rm req}$ is the delivered fraction needed to match the observed lithium with outbursts included (an upper limit for XTE J1118+480). The last column gives the delivered beryllium as A(Be) when the model reproduces the observed lithium; the solar value is 1.38 [24].

System	$f_{\rm int}$ (SERPENS)	$f_{\rm req}$	A(Li), quiescent	A(Li), + outbursts	observed A(Li)	predicted A(Be)
A0620−00	1.5×10^{-4}	4×10^{-5}–4×10^{-3}	0.1 to 2.1	0.9 to 2.9	2.31 ± 0.21	1.6
XTE J1118+480	0.9×10^{-4}	$\leq 9\times10^{-6}$–9×10^{-4}	−0.3 to 1.7	0.8 to 2.8	≤ 1.78	≤ 1.1
V404 Cyg	1.5×10^{-4}	6×10^{-6}–6×10^{-4}	2.1 to 4.1	2.1 to 4.1	2.70 ± 0.40	2.1
Gaia BH1	5×10^{-8}	–	−18 to −16	–	2.3 ± 0.1 (intrinsic)	–
Gaia BH2	2.5×10^{-7}	–	−14 to −12	–	not detected	–
Gaia BH3	9×10^{-9}	–	−13 to −11	–	normal	–

giant and 1.4×10^{-3} for Gaia BH3, giving $f_{\rm int}$ between 10^{-8} and 3×10^{-7} (Figure 5b). The delivered lithium comes to $A({\rm Li}) \approx -17$ for Gaia BH1 and −11 to −14 for the two giants (Table III), at least ten orders of magnitude below anything measurable.

The observations agree. The Gaia BH1 companion does have lithium, $A({\rm Li}) = 2.3\pm0.1$, but El-Badry et al. [12] show that this is ordinary for a solar-type star older than the Hyades; our model attributes none of it to the black hole. The Gaia BH2 giant shows no lithium, as expected after first dredge-up [13], and the Gaia BH3 giant has no peculiarity in 29 elements from lithium to thorium [15]. The Gaia BH2 companion is α-enhanced, $[\alpha/{\rm Fe}] = +0.26\pm0.05$, despite a thin-disk orbit [13]. Spallation cannot produce this: it makes Li, Be and B and destroys, rather than adds, oxygen. El-Badry et al. show that direct capture of supernova ejecta would deliver at most $7\times10^{-6}\,M_\odot$ of oxygen at this separation, too little to matter, and suggest that slow, bound ejecta might work if modelled in detail. Either way, the α-enhancement and the lithium probe different processes, and a lithium or beryllium excess in a Gaia black hole companion would point to something other than the hot-flow mechanism studied here.

E. A beryllium test

The same flow makes beryllium from oxygen. In the linear regime Be/Li in a processed parcel is fixed by the ratio of the proton and α fluences, and the delivered beryllium comes out 0.6–0.7 dex below the delivered lithium (Table III). If the model reproduces the observed lithium, the beryllium delivered to the V404 Cyg companion corresponds to $A(\mathrm{Be}) \approx 2.1$, five times the solar value. For A0620−00 the delivered beryllium, $A(\mathrm{Be}) \approx 1.6$, is comparable to what the star started with, so the surface abundance would rise by a factor of about two; for XTE J1118+480 the delivered amount is below solar and would not be detectable. Beryllium burns at a higher temperature than lithium, so its depletion time is longer and these numbers are lower limits within the model. The Be II resonance doublet at 3130–3131 Å is accessible from the ground for the brighter companions and is the direct analogue of the beryllium measurement that started the polluted white dwarf argument.

V. DISCUSSION

A. What the agreement does and does not show

The model has three parts. The dynamics give f_{esc} and g; this part is checked against exact geodesics and we trust it to the Monte Carlo error of a few percent. The geometry gives θ, which is fixed by the measured mass ratios. The weak parts are the hot-flow microphysics and the companion. Table IV shows how the fiducial prediction ($r_0 = 10\,r_s$, outbursts included, $\tau_{\mathrm{dep}} = 10^8$ yr) moves when one input changes.

The ion temperature is the largest single uncertainty. The lithium-producing α particles sit in the exponential tail above 35 MeV, so raising T_i from 55% to 100% of virial adds 1.5 dex. Real hot flows also lose mass in winds [6], which would raise f_{esc} above our single-attempt ballistic value. Both effects push the prediction up. Since the fiducial model already matches the data, a hotter flow or a strong wind would require depletion times shorter than those quoted above to stay consistent. The agreement in Figure 5 is therefore an order-of-magnitude statement: the mechanism delivers the right amount of lithium for plausible inputs, without tuning. It does not pin down any single parameter.

B. Limitations

Our ions are ballistic once launched. In a real ADAF the magnetic field binds the ions to the flow, and escape happens through winds and jets whose launching we do not model. The escape fraction of 1.5% should be read as the processed fraction available to an outflow. We also assume the companion intercepts the outflow geometrically, ignore the companion's own wind and magnetosphere, and treat the outbursts with exponential light curves rather than the observed ones. The lithium depletion time is the least constrained quantity in the problem and spans two decades. Finally, we have not modelled lithium made during the bright soft states, which Yi & Narayan [11] argued could add to the total; including it would strengthen rather than weaken the case.

TABLE IV: Change in predicted $A(\mathrm{Li})$ when one input changes, at $r_0 = 10\,r_s$ with outbursts and $\tau_{\mathrm{dep}} = 10^8$ yr.

Change	$\Delta A(\mathrm{Li})$ [dex]
kT_i: 0.55 → 1.0 of virial	+1.5
σ_{Li} (α+α) ×2	+0.2
τ_{dep}: $10^8 \rightarrow 10^9$ yr	+1.0
$f_{\mathrm{esc}} \times 10$ (MHD winds)	+1.0
$M_{\mathrm{conv}} \times 2$	−0.3
r_0: $10 \rightarrow 5\,r_s$	+0.8
r_0: $10 \rightarrow 20\,r_s$	−0.9

C. Relation to other explanations

Two other origins have been proposed for the lithium. Lithium could have been made in the supernova that formed the black hole and captured by the companion, as has been argued for the α-element excesses in some of these systems [10, 37]. But a K or M dwarf with a deep convection zone would have destroyed that lithium over the $\gtrsim 10^8$ yr since the explosion. Alternatively, the systems could be young enough to retain primordial lithium, which González Hernández et al. [10] note would make A0620−00 as young as the Pleiades. Ongoing spallation in the hot flow avoids both problems because it resupplies the lithium continuously. The beryllium prediction distinguishes it from the primordial explanation: a young star would show solar beryllium, not the fivefold excess we predict for V404 Cyg.

D. The lithium isotope ratio

The ratio $^6\mathrm{Li}/^7\mathrm{Li}$ has been proposed as a way to tell fresh production from preserved lithium. Casares et al. [9] measured $^6\mathrm{Li}/^7\mathrm{Li} = 0.12^{+0.08}_{-0.05}$ in the neutron star binary Cen X-4 and read the low ^{6}Li as favouring preservation, though they did not rule out production. Two points weaken the test for our model. First, the production ratio itself depends on the ion temperature: $\alpha + \alpha$ reactions in a hot flow give $0.1 < {}^6\mathrm{Li}/{}^7\mathrm{Li} < 10$ for kT_i between 3 and 50 MeV, and the Cen X-4 value corresponds to $kT_i \approx 4$ MeV [9], inside the 4–20 MeV range of our flow. Second, the companion destroys ^{6}Li at lower temperatures than ^{7}Li [49]. In steady state each isotope builds up to its delivery rate times its own depletion time, so the surface ratio is the production ratio multiplied by $\tau_{\mathrm{dep}}(^6\mathrm{Li})/\tau_{\mathrm{dep}}(^7\mathrm{Li}) < 1$. Freshly delivered lithium should therefore look ^{6}Li-poor at the surface, much like preserved lithium. We do not model the isotopic cross sections or the companion's burning in enough detail to predict the ratio, so we treat it as non-discriminating. Beryllium does discriminate, as noted above.

Casares et al. also note that lithium has only been detected in companions earlier than K5 and never above the meteoritic $A(\mathrm{Li}) = 3.3$ [9]. Both facts fit our model as well as they fit preservation. Later spectral types have deeper convection zones and shorter depletion times, which lowers the delivered lithium through equation (11), and the depletion times we infer, $\lesssim 3 \times 10^8$ yr, keep the steady-state abundance below the meteoritic value.

E. Future work

The most useful next step is observational: near-UV spectra of the V404 Cyg companion to measure beryllium. On the modelling side, the escape fraction should be recomputed with an MHD wind launched from the hot flow, and the companion's response should include rotational mixing, which sets τ_{dep}. A spinning black hole moves the ISCO inward and raises the ion temperatures near it; with Kerr geodesics in place of equation (8) the same calculation would test whether the lithium excess correlates with spin. Gaia DR4 is expected to deliver many more dormant black holes [12]; our model predicts that none of their companions will carry spallation lithium.

VI. CONCLUSIONS

We have carried the polluted white dwarf spallation argument of Doyle et al. [1] over to black holes, using SERPENS to follow ions evaporated from the hot accretion flow. Our main results are:

1. About 1.5% of the ions evaporated from the hot flow escape; the rest cross the ISCO. Exact Schwarzschild geodesics and Newtonian gravity give the same escape fraction, while the 1PN approximation in REBOUNDx overestimates it by a factor of 3.7 at $10\,r_s$ and fails to converge further in.

2. A parcel of gas falling through the hot flow of a quiescent black hole collects in about 40 s the proton fluence that Doyle et al. attribute to $\sim 10^7$ yr in a Jovian radiation belt.

3. The fraction of the accreted gas that reaches the companion is $0.9\text{–}1.5 \times 10^{-4}$ in A0620−00, XTE J1118+480 and V404 Cyg. This is the free parameter of Yi & Narayan's model, and it falls inside the range they required.

4. With published ADAF accretion rates and the observed outburst histories, the predicted companion lithium matches the measured values in A0620−00 and V404 Cyg and respects the upper limit in XTE J1118+480, for depletion times of 4×10^7 yr (subgiant) to 3×10^8 yr (K dwarf).

5. The dormant Gaia black holes deliver at least ten orders of magnitude too little lithium to be seen. This agrees with the measured companions: the lithium in the Gaia BH1 G dwarf is normal for its age, the Gaia BH2 giant shows none, and the Gaia BH3 giant has no peculiarities.

6. The model predicts a beryllium abundance about five times solar in the V404 Cyg companion, which near-UV spectroscopy can test. The lithium isotope ratio does not separate production from preservation, because stars destroy ^{6}Li before ^{7}Li; beryllium does.

Appendix A: Integrator benchmark at binary scales

Before the strong-field work, we tested the GR-extended SERPENS integrator on a benchmark system, `StellarBH-10`: a $10\,M_\odot$ black hole with a mass-losing source on a near-circular orbit at $a = 5 \times 10^8$ m ($1.7 \times 10^4\, r_s$). Hydrogen is sputtered from the source with the Smyth–Combi distribution [50], with bulk and maximum speeds of 5% and 30% of the local orbital speed. We integrated eight source orbits with the Wisdom–Holman scheme [51] and the REBOUNDx `gr_full` force [20, 52]. At these separations $v/c \sim 10^{-3}$ and the 1PN expansion is accurate. The particles form a ring at the source orbit (Figure 6) and none reach the ISCO. Over two orbits of the Gaia BH1 configuration the relative energy error is $|\Delta E/E| \simeq 3.6 \times 10^{-9}$. These tests show that the integrator is sound at binary scales; Section IV A shows where the 1PN force stops being adequate.

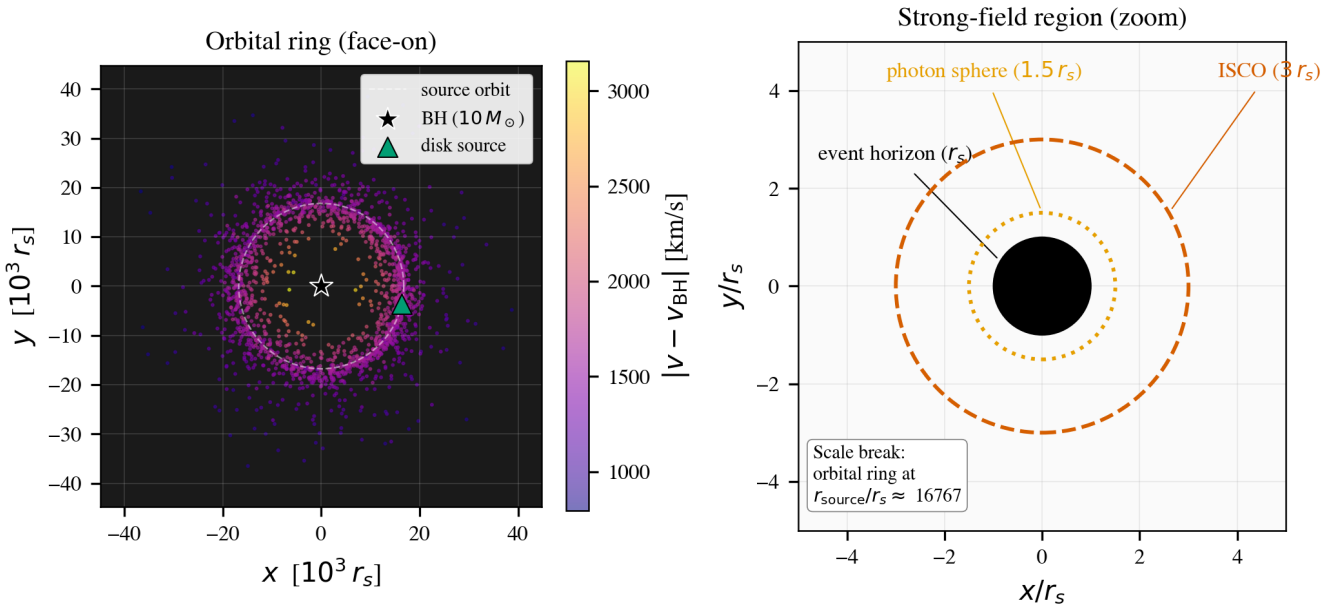


FIG. 6: Particle distribution in the orbital plane of the `StellarBH-10` benchmark. *Left:* full view, with the source (green triangle) and the ring of ~ 1900 test particles coloured by speed relative to the black hole. *Right:* the strong-field region, showing the horizon at r_s (filled), the photon sphere at $1.5\,r_s$ (dotted) and the ISCO at $3\,r_s$ (dashed red) [16]. Distances in units of r_s.

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